UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 005 · Homogeneous functions and Euler's theorem · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S005·6	Deck 005, slide 6 (rendered .pptx)
T005 00:08:57	AV-summary transcript 005, segment start
CW U1 P20	Unit 1 class-work problem 20
ESA Jan-24 s3 1c	Past paper: Jan 2024, set 3, question 1.c

Contents

1. Figure G1.1 · Scaling along a ray: $u(tx, ty) = t^n u(x, y)$
2. Figure G2.1 · Level curves of a homogeneous function are scaled copies

Colours follow the Okabe–Ito order. Both figures are added for clarity, since the deck has no graphs: one shows how a homogeneous function scales along a ray, the other shows level curves of $x^2 + xy + y^2$ as scaled copies.

1 Homogeneous functions

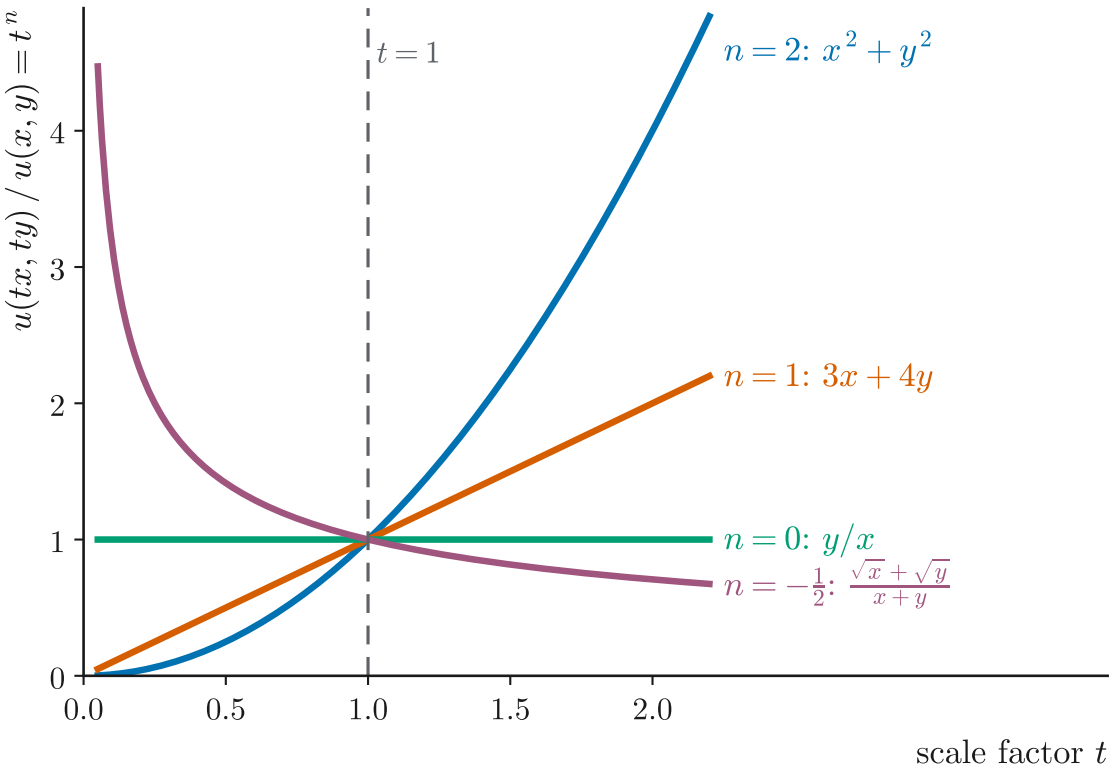


Figure G1.1. Scaling along a ray: $u(tx, ty) = t^n u(x, y)$. ESA ★★

ADDED FOR CLARITY S005·3 · S005·5

How to read it. Start at a point and move out along the ray through the origin by the factor t. A homogeneous function of degree n changes like t^n: degree 2 grows quadratically, degree 1 linearly, degree 0 stays constant along the ray (but can vary between rays), and degree $-\frac{1}{2}$ decays. Each curve is $u(t)/u(1)$.

2 Euler's theorem

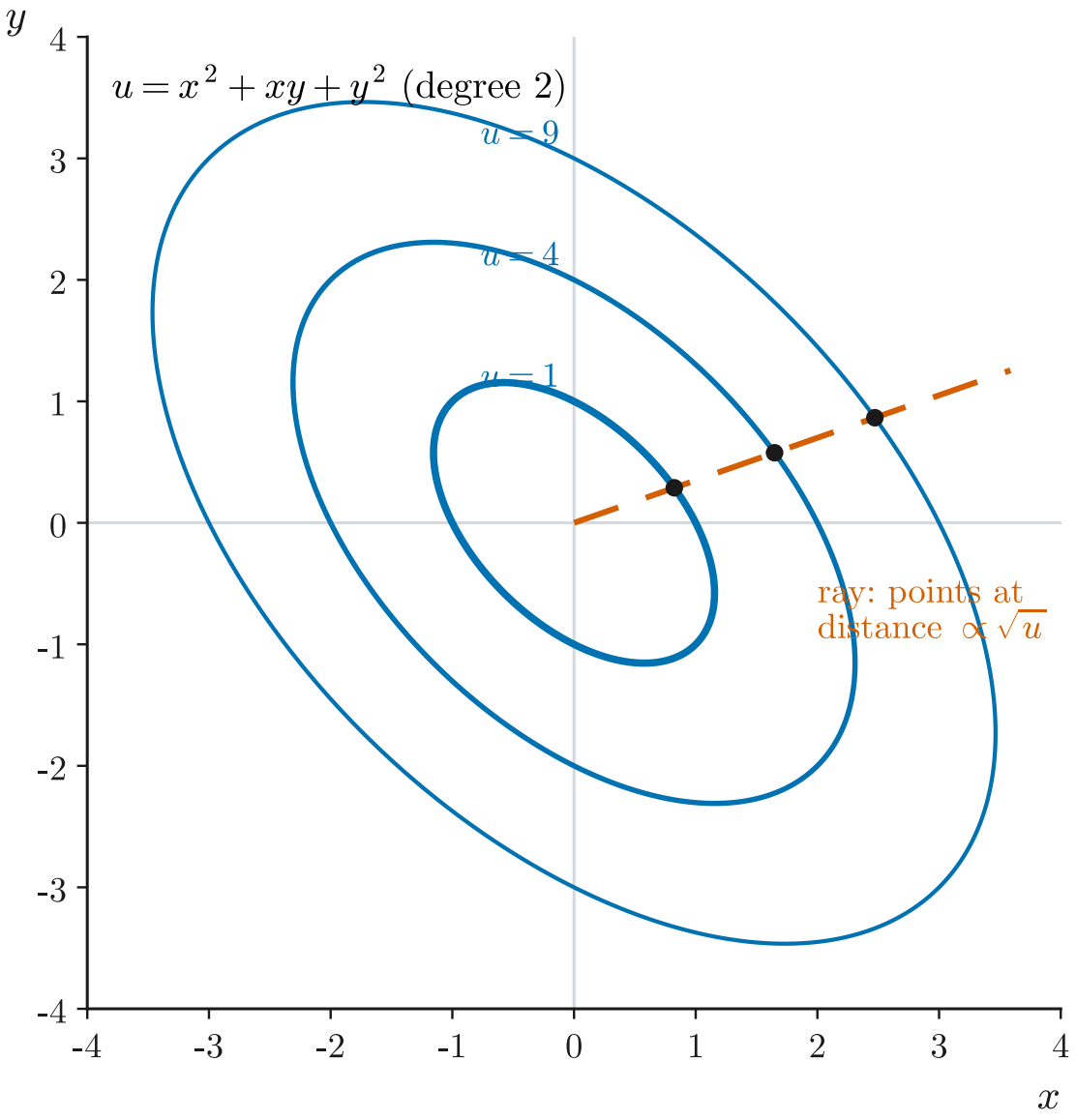


Figure G2.1. Level curves of a homogeneous function are scaled copies. ESA ★★

ADDED FOR CLARITY S005·5 · T005 00:10:18

How to read it. Level curves of $u = x^2 + xy + y^2$ (degree 2) for $u = 1, 4, 9$. Each is the first one scaled by $1, 2, 3 = \sqrt{u}$. Moving out along the dashed ray, the rate of change $xu_x + yu_y$ (the radial change scaled by distance) equals $2u$: this is Euler's theorem.